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Exercise 9

Due date: **Tuesday, May 19th 2026, 14:00**

Submission Notes:

- Please submit your answers using the corresponding function in ADAM.
- Write down your solutions in a way that your reasoning and solution path is understandable.
- Please submit your solutions to the coding tasks as executable code and make sure that your implementations are working. If you encounter problems and cannot solve them, add additional notes and comments to your solution about where you think the error is or what you have tried and tested so far.
- Plagiarism results in an automatic fail for exercise sheets, i.e. every student needs to hand in their own solution.

9.1 Gaussian Mixture Model

For now, we discussed classification and regression problems, where the model captures the relationship between an input $\mathbf{x}$ and a label y . However, a modification of the classification scenario could involve data with no labels or additional unobserved clusters. In order to identify the missing labels, one can use a mixture models approach. A Gaussian mixture model assumes that the data is Gaussian distributed. Given d -dimensional observations $(x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, K unidentified clusters and corresponding mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$ parameters for each k -th multivariate Gaussian distribution, the model parameters can be estimated with the EM-algorithm.

Definitions

- The **Gaussian mixture model** has the form

$$p(\mathbf{x}_i | \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

where K indicates the amount of clusters, π_k assigns probability for the k -th cluster and $\boldsymbol{\theta}_k = (\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ parametrizes the k -th multivariate Gaussian distribution.

- The **expected complete data log-likelihood** is given by

$$\begin{aligned}
Q(\boldsymbol{\theta}, \boldsymbol{\theta}^t) &\equiv \mathbb{E} \left[\sum_i \log p(\mathbf{x}_i, z_i \mid \boldsymbol{\theta}) \right] = \\
&= \sum_i \mathbb{E} \left[\log \left(\prod_{k=1}^K \left(\pi_k p(\mathbf{x}_i \mid \boldsymbol{\theta}_k) \right)^{\delta(z_i=k)} \right) \right] = \\
&= \sum_i \sum_k \mathbb{E} [\delta(z_i = k)] \log [\pi_k p(\mathbf{x}_i \mid \boldsymbol{\theta}_k)] = \\
&= \sum_i \sum_k p(z_i = k \mid \mathbf{x}_i, \boldsymbol{\theta}^t) \log [\pi_k p(\mathbf{x}_i \mid \boldsymbol{\theta}_k)] = \\
&= \sum_i \sum_k r_{ik}^t \log \pi_k + \sum_i \sum_k r_{ik}^t \log p(\mathbf{x}_i \mid \boldsymbol{\theta}_k),
\end{aligned}$$

where $\delta(z_i = k)$ assigns a so called hard label, $r_{ik}^t \triangleq p(z_i = k \mid \mathbf{x}_i, \boldsymbol{\theta}^t)$ assigns a soft label as a probability value that a data point $\mathbf{x}_i$ was generated from the k -th distribution. Note that the Q -function is also called an *auxiliary function*.

- The **E-step** has the following form

$$r_{ik}^t = \frac{\pi_k p(\mathbf{x}_i \mid \boldsymbol{\theta}_k^t)}{\sum_{k'} \pi_{k'} p(\mathbf{x}_i \mid \boldsymbol{\theta}_{k'}^t)},$$

where

$$\pi_k = \frac{1}{N} \sum_i r_{ik}^t.$$

Both r_{ik}^t and π_k are updated in each iteration step.

- The **M-step** involves a following optimization task

$$\boldsymbol{\theta}^{t+1} = \arg \max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^t),$$

where the auxiliary function Q is optimized w.r.t. the model parameters $\boldsymbol{\theta}$ and the soft labels. The new parameter estimates at the iteration step t are given by

$$\begin{aligned}
\boldsymbol{\mu}_k &= \frac{\sum_i r_{ik}^t \mathbf{x}_i}{\sum_i r_{ik}^t}, \\
\boldsymbol{\Sigma}_k &= \frac{\sum_i r_{ik}^t (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top}{\sum_i r_{ik}^t}.
\end{aligned}$$

Exercise

Derive the parameter estimates $\boldsymbol{\theta}_k = (\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ given the auxiliary function Q . Is the convergence to a global optimum guaranteed?

9.2 Python Implementation

In the results derived above, the E- step and the M-step are defined. Complete the provided template `Exercise9.py` to fit a mixture of Gaussians using the EM-algorithm.

- Generate a dataset from two independent 2-dimensional Gaussians.

- Compute in the E-step the soft labels, i.e. the probability that a data point is assigned to a particular cluster.
- Compute in the M-step new parameter estimates of multivariate Gaussians.
- Compute the log-likelihood given the model parameters, in order to determine convergence of the algorithm.
- Plot the fitted densities and the observed data points.